CIT
Technical University of Munich

Ecorrection
Place student sticker here

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
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Diskrete Wahrscheinlichkeitstheorie

Exam:	IN0018 / Endterm	**Date:**	Wednesday 5th August, 2026
Examiner:	Niki Kilbertus	**Time:**	08:00 – 10:00

Sample Solution
Correction Notes

Working instructions

- This exam consists of **12 pages** with a total of **5 problems**.
  Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 44 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one **single sheet of A4 paper** with handwritten notes on both sides
  - one **analog dictionary** English ↔ native language **without notes**
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.
- Place your student ID visibly on your table.

Left room from _____ to _____ / Early submission at _____

Problem 1 True or false (5 credits)

For the following statements, mark exactly one answer. No justification is required.

a) Let $X \sim \text{Ber}(p)$ and set $Y := 1 - X$. If $p = 1/2$, then X and Y are uncorrelated.

- ☐ True
- ☒ False

b) For any two Bernoulli RVRVs X and Y, $\text{cov}[X, Y] = 0$ implies $X \perp\!\!\!\perp Y$.

- ☐ False
- ☒ True

c) Let X and Y be independent RVRVs with $\phi_X(t) = \phi_Y(t) = \cos(t)$ for all $t \in \mathbb{R}$. Then

$$\phi_{2X+Y}(t) = \cos(2t)\cos(t)$$

for all $t \in \mathbb{R}$.

- ☐ False
- ☒ True

d) Let $X_1, X_2, \ldots$ be a sequence of RVRVs on a common probability space, and let $c \in \mathbb{R}$. If $X_n \xrightarrow{d} c$, then $X_n \xrightarrow{P} c$.

- ☐ False
- ☒ True

e) Let $(\mathbb{R}, \mathcal{B}, P_\Theta)$ be a statistical model. Let $\hat{\theta}_n^{(\text{mle})}$ be the maximum likelihood estimator for θ and let $\hat{\theta}_n$ be any other estimator for θ. Then, for every finite n and every $\theta \in \Theta$,

$$\text{mse}[\hat{\theta}_n^{(\text{mle})}] \leq \text{mse}[\hat{\theta}_n] \,.$$

- ☒ False
- ☐ True

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Problem 2 True or false with justification (12 credits)

Decide whether each statement is true or false and provide a justification (proof, counterexample, or explanation via known results).

a) Let $X_1, X_2, \ldots$ and X be RVRVs on a common probability space such that $X_n \xrightarrow{d} X$. If all expectations $\mathbb{E}[X_n]$ and $\mathbb{E}[X]$ exist and are finite, then $\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X]$.

0 1 2 3

False. For a counterexample, let $U \sim \mathrm{Unif}(0,1)$ and define

$$X_n := n\chi_{\{U \le 1/n\}}, \qquad X := 0 .$$

These RVRVs are defined on a common probability space and satisfy

$$P(X_n = n) = \frac{1}{n}, \qquad P(X_n = 0) = 1 - \frac{1}{n},$$

while $X = 0$ with probability 1. Then $X_n \xrightarrow{P} 0$, because for every $\epsilon > 0$ and all sufficiently large n,

$$P(|X_n| \ge \epsilon) = \frac{1}{n} \to 0 .$$

Thus $X_n \xrightarrow{d} X$. However, $\mathbb{E}[X_n] = 1$ for every n, whereas $\mathbb{E}[X] = 0$.
1 point for false; 2 points for a valid sequence for the counterexample.

b) Let Ω, Ω' be sets, $\mathcal{A}$ a σ-algebra on Ω, and $f : \Omega \to \Omega'$ an arbitrary function. Then $f(\mathcal{A}) := \{f(A) \mid A \in \mathcal{A}\}$ is a σ-algebra on Ω'.

0 1 2 3

False. For example, let

$$\Omega = \{1,2,3\}, \qquad \Omega' = \{a,b\},$$

let $f(1) = f(2) = a$ and $f(3) = b$, and consider the σ-algebra

$$\mathcal{A} = \{\emptyset, \{1,3\}, \{2\}, \Omega\} .$$

Although f is surjective, its direct images form

$$f(\mathcal{A}) = \{\emptyset, \{a\}, \{a,b\}\} .$$

This is not a σ-algebra on Ω', because it contains $\{a\}$ but not its complement $\{b\}$.
1 point for false; 2 points for a valid counterexample and identification of a failed σ-algebra property.

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c) Let X and Y be integrable RVRVs. If $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$ then X and Y are independent.

False. Let X be uniformly distributed on $\{-1, 0, 1\}$ and set $Y := X^2$. Then $\mathbb{E}[X] = 0$. If $Y = 0$, then $X = 0$; if $Y = 1$, then conditionally X is uniform on $\{-1, 1\}$. Hence

$$\mathbb{E}[X \mid Y = 0] = \mathbb{E}[X \mid Y = 1] = 0,$$

so $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$. Nevertheless, X and Y are not independent. For instance,

$$P(X = 0, Y = 0) = \frac{1}{3} \neq \frac{1}{3} \cdot \frac{1}{3} = P(X = 0)P(Y = 0)\,.$$

1 point for false; 1 point for a valid example satisfying the conditional-expectation identity; 1 point for demonstrating dependence.

0 1 2 3

d) Let X and Y be RVRVs with finite variance. If $X + Y$ and $X - Y$ are independent, then $\mathrm{var}[X] = \mathrm{var}[Y]$.

True. Independence and the finite variances imply

$$\mathrm{cov}[X + Y, X - Y] = 0\,.$$

By bilinearity and symmetry of covariance,

$$\begin{aligned}\mathrm{cov}[X + Y, X - Y] &= \mathrm{cov}[X, X] - \mathrm{cov}[X, Y] + \mathrm{cov}[Y, X] - \mathrm{cov}[Y, Y] \\ &= \mathrm{var}[X] - \mathrm{var}[Y]\,.\end{aligned}$$

It follows that $\mathrm{var}[X] = \mathrm{var}[Y]$.
1 point for true; 1 point for using independence to obtain zero covariance; 1 point for expanding the covariance and concluding equality of the variances.

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Problem 3 Jensen's inequality (5 credits)

a) State Jensen's inequality with all required assumptions.

0 1 2 3

Let X be an RVRV such that $\mathbb{E}[X]$ is finite. Let $g : \mathbb{R} \to \mathbb{R}$ be a convex function. Then

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X]) .$$

(If g is strictly convex, then equality holds if and only if $X = \mathbb{E}[X]$ with probability 1, i.e., X is a constant. If g is concave, the inequality is reversed, $\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$.)
1 point for X with finite expectation; 1 point for $g : \mathbb{R} \to \mathbb{R}$ convex; 1 point for the correct inequality. The parenthetical additions are not graded.

b) Let X be an RVRV such that $\mathbb{E}[X]$ and $\mathbb{E}[e^X]$ exist and are finite. Show that

$$\mathbb{E}[X] \leq \log\big(\mathbb{E}[e^X]\big) .$$

0 1 2

The function $g : \mathbb{R} \to \mathbb{R}$, $g(x) = e^x$, is convex. Jensen's inequality therefore gives

$$e^{\mathbb{E}[X]} \leq \mathbb{E}[e^X] .$$

Since the logarithm is increasing and $\mathbb{E}[e^X] > 0$, taking logarithms yields

$$\mathbb{E}[X] \leq \log\big(\mathbb{E}[e^X]\big) .$$

1 point for the correct Jensen inequality $e^{\mathbb{E}[X]} \leq \mathbb{E}[e^X]$; 1 point for stating that the logarithm is increasing on $(0, \infty)$, observing that $e^x > 0$ and hence $\mathbb{E}[e^X] > 0$, and taking logarithms to obtain the requested bound.

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Problem 4 Characteristic functions and tail bounds (9 credits)

Let $X_1, X_2, \ldots$ be iid RVRVs with

$$P(X_j = 1) = P(X_j = -1) = \frac{1}{2} \qquad \text{for every } j \in \mathbb{N}_{>0}\,.$$

For $n \in \mathbb{N}_{>0}$ define

$$Z_n := \frac{1}{\sqrt{n}} \sum_{j=1}^{n} X_j\,.$$

0 1

a) Compute the characteristic function $\phi_{X_1}(t)$.
Hint: Euler's formula states that $e^{iu} = \cos(u) + i\sin(u)$.

By definition,

$$\phi_{X_1}(t) = \mathbb{E}[e^{itX_1}] = \frac{1}{2}e^{it} + \frac{1}{2}e^{-it} = \cos(t)\,.$$

1 point for the correct computations; simplifying to $\cos(t)$ is not required to get the point.

0 1 2 3

b) Compute the characteristic function $\phi_{Z_n}(t)$.

Using independence,

$$\begin{aligned}\phi_{Z_n}(t) &= \mathbb{E}\left[\exp\left(it\frac{1}{\sqrt{n}}\sum_{j=1}^{n} X_j\right)\right] \\ &= \prod_{j=1}^{n} \mathbb{E}\left[\exp\left(i\frac{t}{\sqrt{n}}X_j\right)\right] \\ &= \left(\phi_{X_1}\left(\frac{t}{\sqrt{n}}\right)\right)^n = \left(\cos\left(\frac{t}{\sqrt{n}}\right)\right)^n\,.\end{aligned}$$

1 point for scaling by $t/\sqrt{n}$, 1 point for using independence to get a product, 1 point for the final expression. If there was a wrong result in (a), but (b) correctly carries it forward using the correct scaling and n-th power of the function from (a), do not deduct points in (b).

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c) Compute $\mathbb{E}[Z_n]$ and $\mathrm{var}[Z_n]$.

0 1 2

For each j, we have

$$\mathbb{E}[X_j] = \frac{1}{2} - \frac{1}{2} = 0, \qquad \mathrm{var}[X_j] = \mathbb{E}[X_j^2] - \mathbb{E}[X_j]^2 = 1\,.$$

Therefore,

$$\mathbb{E}[Z_n] = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} \mathbb{E}[X_j] = 0\,.$$

By independence,

$$\mathrm{var}[Z_n] = \frac{1}{n} \sum_{j=1}^{n} \mathrm{var}[X_j] = 1\,.$$

Alternatively, using the characteristic function from (b), differentiation at zero gives

$$\phi'_{Z_n}(0) = 0, \qquad \phi''_{Z_n}(0) = -1\,.$$

The moment identities for characteristic functions then yield

$$\mathbb{E}[Z_n] = \frac{1}{i}\phi'_{Z_n}(0) = 0, \qquad \mathbb{E}[Z_n^2] = \frac{1}{i^2}\phi''_{Z_n}(0) = 1,$$

and hence $\mathrm{var}[Z_n] = 1$.
1 point for the mean and 1 point for the variance, whether computed directly or from the characteristic function.

d) Let $a > 0$. Use Markov's inequality to derive an upper bound for $P(|Z_n| \geq a)$. Use the clt to approximate $P(|Z_n| \geq a)$. If you could not solve (c), you may use $\mathbb{E}[Z_n^2] = 1$ here.

0 1 2 3

The random variable Z_n^2 is non-negative. Applying Markov's inequality to Z_n^2 gives

$$P(|Z_n| \geq a) = P(Z_n^2 \geq a^2) \leq \frac{\mathbb{E}[Z_n^2]}{a^2} = \frac{1}{a^2}\,,$$

where we used $\mathbb{E}[Z_n^2] = \mathrm{var}[Z_n] + \mathbb{E}[Z_n]^2 = 1$.
Since the X_j are iid with mean zero and variance one, the clt yields $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. Thus, for large n,

$$P(|Z_n| \geq a) \approx z_{a/2}\,,$$

where $z_{a/2}$ is the $1 - a/2$ quantile of a standard Gaussian.
1 point for the Markov bound $1/a^2$; 1 point for $Z_n \xrightarrow{d} \mathcal{N}(0,1)$; 1 point for the approximation $z_{a/2}$;

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Problem 5 **Randomized response (13 credits)**

In a survey, each participant has a sensitive binary attribute $A \sim \mathrm{Ber}(p)$, where $p \in [0,1]$ is unknown. Instead of observing A directly, we observe a privacy-protected answer Y generated as follows: with probability $r \in (0,1]$, the participant answers truthfully, and with probability $1-r$, the participant answers by flipping a fair coin independently of A. We observe iid answers $Y_1, \dots, Y_n$ generated by this mechanism with known r. Let $\bar{Y}_n := \frac{1}{n}\sum_{i=1}^{n} Y_i$.

0 1 2 3

a) Compute $P(Y=1)$ in terms of p and r. Is p identifiable from the distribution of Y?

By the law of total probability over the two response mechanisms,

$$P(Y=1) = r\,P(A=1) + (1-r)\frac{1}{2}$$
$$= rp + \frac{1-r}{2} =: q(p)\,.$$

Since $r > 0$, the map $p \mapsto q(p) = rp + (1-r)/2$ is injective on $[0,1]$. Thus p is identifiable from the distribution of Y.
1.5 points for the correct expression for $P(Y=1)$, 1 point for the injectivity argument, 0.5 points for the identifiability conclusion.

0 1 2 3

b) Assume that the observed value $\bar{y}_n$ lies in the interval $\left[\frac{1-r}{2}, \frac{1+r}{2}\right]$. Derive the maximum likelihood estimator for p.
Hint: Recall that the maximum likelihood estimator in a Bernoulli model is the sample mean.

The observed answers are iid Bernoulli with success probability

$$q(p) = rp + \frac{1-r}{2}\,.$$

For an observed sample $y = (y_1, \dots, y_n)$ with $\bar{y}_n = n^{-1}\sum_i y_i$, the likelihood as a function of q is

$$L(q \mid y) = q^{\sum_i y_i}(1-q)^{n-\sum_i y_i}\,.$$

The Bernoulli likelihood is maximized at $\hat{q}_{\mathrm{mle}} = \bar{y}_n$. Solving $\bar{y}_n = rp + (1-r)/2$ for p gives

$$\hat{p}_{\mathrm{mle}} = \frac{\bar{Y}_n - \frac{1-r}{2}}{r}\,.$$

The assumption on $\bar{y}_n$ ensures that this value lies in $[0,1]$.
1 point for the Bernoulli likelihood of the observed data, 1 point for $\hat{q}_{\mathrm{mle}} = \bar{Y}_n$, 1 point for transforming back to $\hat{p}_{\mathrm{mle}}$.

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c) Show that

$$\hat{p} := \frac{\bar{Y}_n - \frac{1-r}{2}}{r}$$

is unbiased for p and compute $\mathrm{var}[\hat{p}]$.

0 1 2

Since $Y_i \sim \mathrm{Ber}(q(p))$ with $q(p) = rp + (1-r)/2$,

$$\mathbb{E}[\hat{p}] = \frac{\mathbb{E}[\bar{Y}_n] - \frac{1-r}{2}}{r} = \frac{q(p) - \frac{1-r}{2}}{r} = p\,.$$

Thus $\hat{p}$ is unbiased. Moreover,

$$\mathrm{var}[\hat{p}] = \frac{1}{r^2}\,\mathrm{var}[\bar{Y}_n] = \frac{1}{r^2}\,\frac{q(p)(1-q(p))}{n}\,.$$

1 point for unbiasedness, 1 point for the variance.

d) The survey organizers want to determine whether more than 60% of the population have the sensitive attribute. For this subproblem, suppose that $r = 1/2$. Among $n = 100$ iid responses, 64 responses equal 1. Formulate the hypotheses and perform an approximate test at significance level $\alpha = 0.05$. State the test statistic, its approximate distribution at the null boundary, the rejection rule, the observed value of the test statistic, and the conclusion.
For this subproblem, you may use $q(p) = rp + \frac{1-r}{2}$ and $z_{0.05} \approx 1.64$.

0 1 2 3 4 5

The hypotheses are

$$H_0 : p \le 0.6 \qquad \text{against} \qquad H_1 : p > 0.6\,.$$

Since $q(p)$ is increasing in p, the null boundary is $p_0 = 0.6$, where

$$q_0 = q(0.6) = \frac{1}{2} \cdot 0.6 + \frac{1}{4} = 0.55\,.$$

At this boundary, $\mathbb{E}[\bar{Y}_{100}] = 0.55$ and

$$\mathrm{var}[\bar{Y}_{100}] = \frac{0.55(1-0.55)}{100}\,.$$

By the clt, the test statistic

$$Z := \frac{\bar{Y}_{100} - 0.55}{\sqrt{0.55(1-0.55)/100}}$$

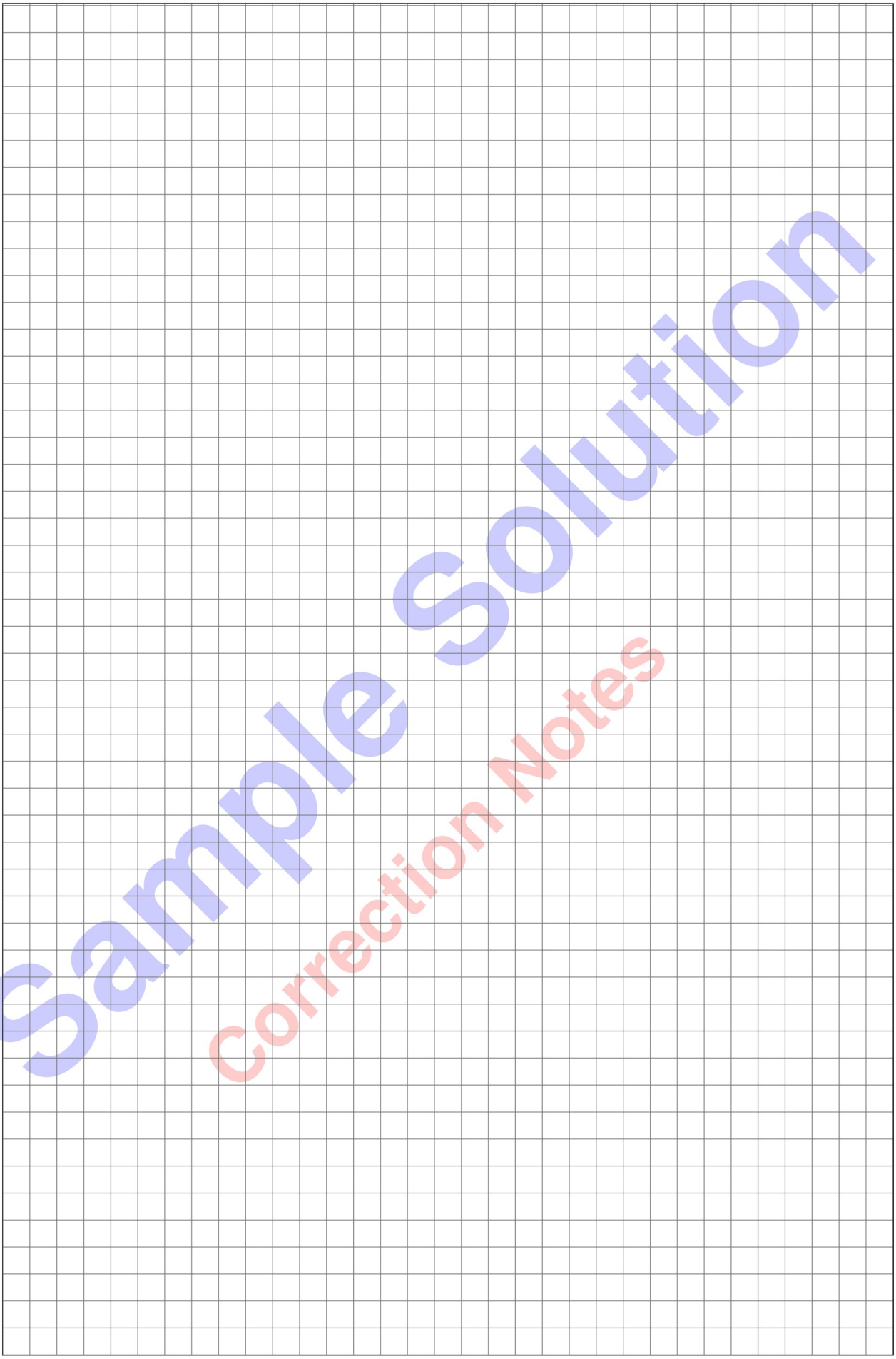
is approximately standard Gaussian under the null boundary. For this one-sided test, we reject H_0 if $z_{\mathrm{obs}} > z_{0.05} \approx 1.64$. The observed sample mean is $\bar{y}_{100} = 0.64$, and hence

$$z_{\mathrm{obs}} = \frac{0.64 - 0.55}{\sqrt{0.55 \cdot 0.45/100}} \approx 1.81\,.$$

Since $1.81 > 1.64$, we reject H_0 at the approximate significance level 0.05. The data provide evidence that $p > 0.6$.
1 point for the correct one-sided hypotheses; 1 point for computing the null-boundary probability $q_0 = 0.55$ and the corresponding mean and variance of $\bar{Y}_{100}$; 1 point for the correctly standardized statistic and its approximate standard Gaussian distribution under the null boundary; 1 point for the rejection rule and the observed value $z_{\mathrm{obs}} \approx 1.81$; 1 point for the correct rejection decision and conclusion.

Sample Solution
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Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.



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